DataCo Smart Supply Chain Analysis

1. Project Objective

To analyze supply chain efficiency, financial health, and delivery performance. This dashboard identifies risk zones in global shipping and optimizes profit margins by analyzing discounts and late deliveries.The main objective of this project is to perform an end-to-end supply chain data analysis to optimize logistics operations using:

- **Microsoft Excel** for initial data exploration and quality assessment.
- **Power BI** for advanced data modeling, DAX computations, and interactive dashboard creation.

This project focuses on analyzing the DataCo Smart Supply Chain dataset:

To Analyze Global Supply Chain Health: To provide a comprehensive overview of financial and operational performance, including Gross Sales ($36.78M), Net Profit ($3.97M), and overall business stability.

To Quantify and Mitigate Late Delivery Risk: To identify the factors driving the current **55% delivery risk** and pinpoint the specific geographic regions with the highest failure rates.

To Conduct Granular Financial Leakage Analysis: To perform a deep-dive into "Loss Zones" where excessive discounts and high shipping costs are eroding profit margins, specifically within the **Consumer** segment.

To Evaluate Shipping Mode Efficiency: To monitor the performance of various shipping methods and investigate why priority modes like **'First Class'** show a **1.8x higher risk** of delay compared to standard modes.

To Address Operational Lag: To bridge the gap between the actual average delivery time **(7 days)** and the corporate target **(4 days)** by identifying process bottlenecks.

To Perform AI-Driven Root Cause Analysis: To utilize **Decomposition Trees** and **Key Influencers** to diagnose the primary drivers of logistical bottlenecks across high-volume categories like **Technology**.

To Map Geographical Risk Hotspots:To visualize and isolate regions such as **Western Europe and Central America** where high sales volume coincides with peak late delivery rates for targeted intervention.

2. Data Sources

- **Dataset:** DataCo Smart Supply Chain (CSV format).
- **Source :** https://data.mendeley.com/datasets/8gx2fvg2k6/5
- **Timeline:** 2019
- **Domain:** Logistics and Supply Chain

3. Problem Statement

Inefficiencies in supply chain operations often lead to late deliveries, which increase operational costs and reduce customer satisfaction. The problems addressed in this project include:

- **Logistical Bottlenecks:** Evaluating the primary drivers of **Late Delivery Risk** (currently 55%) across global territories.
- **Mode Inconsistency:** Determining why priority shipping modes, such as **First Class**, fail to meet scheduled delivery timelines.
- **Financial Leakage:** Identifying "Hotspot" regions where high sales volume coincides with **negative profit margins** due to excessive discounting or shipping variances.
- **Visibility Gap:** Developing a professional, **real-time 3D Dashboard** to replace manual reporting and provide immediate visibility into supply chain KPIs.

4.Attributes details

Attribute Name	Data Type	Description
Type	Category	Payment method used (DEBIT, CASH, PAYMENT, etc.).
Days for shipping (real)	Integer	Actual number of days taken to ship the order.
Days for shipment (scheduled)	Integer	Target or promised number of days for the shipment.
Delivery Status	Category	Status (Late delivery, Advance shipping, Shipping on time).
Late_delivery_risk	Binary	1 if late/at risk, 0 otherwise.
Customer Id	Integer	Unique identifier for each customer.
Customer Name	Text	Combined First and Last name of the customer.

Customer Country	Text	Country of residence for the customer.
Customer State	Text	State of residence for the customer.
Customer City	Text	City of residence for the customer.
Customer Street	Text	Street address of the customer.
Customer Segment	Category	Market segment (Consumer, Corporate, Home Office).
Customer Zipcode	Text	Postal code of the customer.
Latitude	Decimal	Geographic latitude of the customer's location.
Longitude	Decimal	Geographic longitude of the customer's location.
Market	Category	High-level market region (e.g., Pacific Asia, Europe).
Order Id	Integer	Unique identifier for the specific order.
Order Country	Text	Destination country of the order.
Order City	Text	Destination city of the order.
Order date (DateOrders)	Date/Time	Timestamp of when the order was placed.
Department Id	Integer	Unique ID of the department managing the product.
Department Name	Text	Name of the department (e.g., Fitness, Apparel).
Category Id	Integer	Unique ID of the product category.
Category Name	Text	Name of the product category.
Product Card Id	Integer	Unique ID associated with the product card.
Product Name	Text	Official name of the product sold.
Order Item Id	Integer	Unique identifier for each line item within an order.
Order Item Product Price	Decimal	Unit price of the individual product.
Order Item Quantity	Integer	Number of units purchased.
Sales	Decimal	Gross sales value (Price x Quantity).
Order Item Discount	Decimal	Total discount amount applied.
Sales per customer	Decimal	Net revenue (Sales minus Discount).
Order Profit Per Order	Decimal	Net profit generated from the specific order item.
Order Item Discount Rate	Decimal	The percentage of discount applied.
Order Item Profit Ratio	Decimal	The ratio of profit to the total sales value.
Order Region	Text	Specific sub-region where the order was placed.
Order State	Text	Specific state or province of the destination.

Order Status	Category	Current state (COMPLETE, PENDING, FRAUD, etc.).
Product Status	Binary	Indicates if the product is active or discontinued.
Shipping date (DateOrders)	Date/Time	Timestamp of when the shipment was processed.
Shipping Mode	Category	Shipping method (Standard, First Class, Same Day, etc.).

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5. Tools & Technologies

- **Microsoft Excel:** Used for data cleaning, handling missing values, and standardizing date formats.
- **Power BI:** Used for star schema data modeling, creating DAX measures, and advanced 3D visual layouts.
- **Powerquery Editor-** Used for Data cleaning and Data transformation
- **GitHub:** For hosting the project portfolio and sharing findings.

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6. Data Pre-Processing

6.1 Data Imputation & Auditing (Excel)

- **Dimensionality Reduction:** Audited the raw dataset and removed redundant or empty columns, streamlining the data from 53 initial attributes to 41 essential features.
- **Missing Value Imputation:** Identified and filled missing entries in the "Customer Zipcode" field to ensure geographic data integrity.
- **Reference Table Creation:** Developed custom State and Country Mapping tables within Excel to facilitate standardized reporting.

6.2 Data Transformation (Power Query)

- **Geographic Standardization:** Standardized location attributes (City, State, and Country) by translating Spanish nomenclature into English for global consistency.
- **Calculated Attributes:**
    - **Delivery Delay:** Engineered a new feature by calculating the variance between "Days for Shipping (Real)" and "Days for Shipment (Scheduled)."

    - **Arrival Logic:** Derived the "Delivery Date" by adding "Days for Shipping (Real)" to the original "Shipping Date."

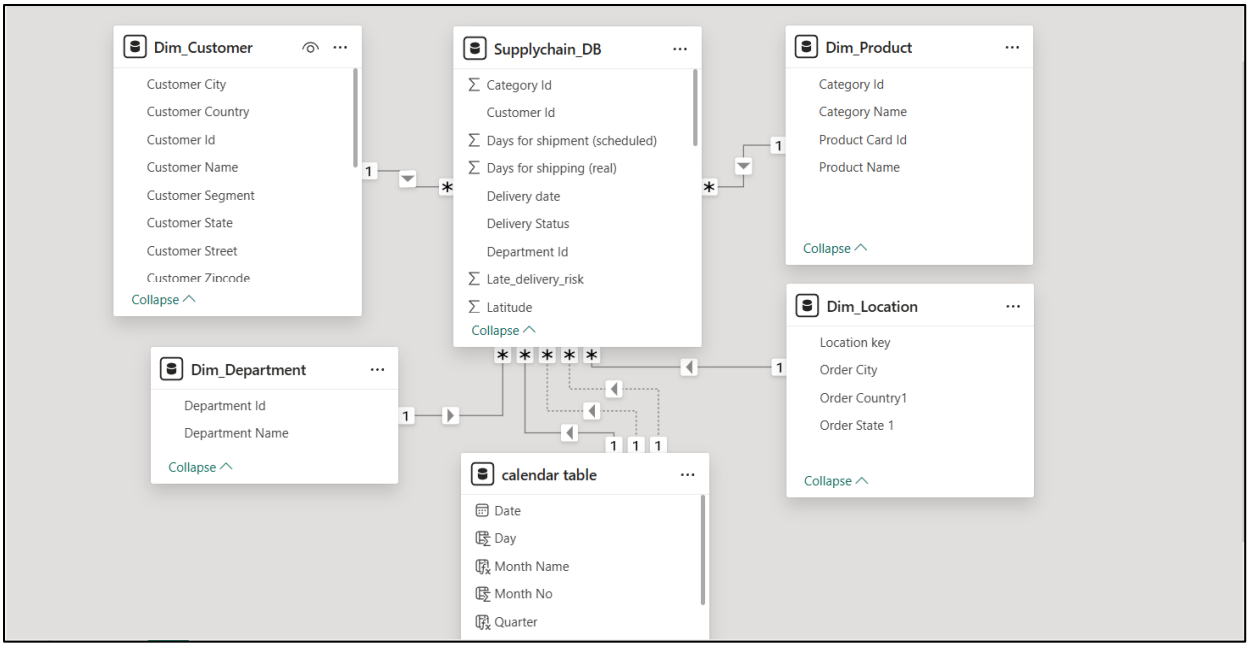
- **Data Integration & Merging:**
    - **Location Key Generation:** Utilized the "Merge Columns" feature to combine Order Country, City, and State into a unique **Location Key** for accurate mapping.
    - **Star Schema Preparation:** Employed **Merge Queries** to consolidate data and generate specialized dimension tables for Products, Customers, Location, and Departments.

- **Data Quality Assurance:**
    - Used **Column Distribution** and **Column Quality** tools to identify duplicates and eliminate hidden empty cells.
    - Validated and updated Data Types for every column to ensure DAX compatibility.
    - Performed final column reordering for a cleaner data model structure.

- **Temporal Modeling:** Developed a dedicated **Calendar Table** using Power BI DAX to enable time-intelligence analysis.

6.3 Data Modeling

- **Fact Table:** Order_Transactions — Acts as the central repository for quantitative metrics, including sales, profit, and delivery performance.
- **Dimension Tables:** Structured the model using Product_Dim, Customer_Dim, Location_Dim, Department_Dim, and a custom Calendar_Table.
- **Schema Architecture:** Implemented a **Star Schema** with **One-to-Many relationships**. This design optimizes query performance and allows for seamless filtering across multiple business dimensions.



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7. DAX Measures & Logic Development

To drive the analytical depth of this project, I developed several custom DAX measures. These formulas allow for dynamic filtering and real-time calculation of key supply chain performance indicators.

7.1. Financial Metrics

These measures track the revenue health and profitability of the global supply chain operations.

- **Gross Sales:**

Gross Sales = SUM(Supplychain_DB[Sales])

- **Total Sales (Net):**

Total Sales = SUM(Supplychain_DB[Sales per customer])

- **Total Profit:**

Total Profit = SUM(Supplychain_DB[Order profit per order])

- **Total Discount:**

Total Discount = SUM(Supplychain_DB[Order Item Discount])

- **Profit Margin %:**

Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)

7.2. Operational & Delivery Metrics

These measures quantify logistical efficiency and the reliability of fulfillment processes.

- **Total Orders:**

Total Orders = COUNT(Supplychain_DB[Order Id])

- **Average Delivery Days:**

Avg Delivery Days = AVERAGEX(Supplychain_DB, DATEDIFF(Supplychain_DB[order date], Supplychain_DB[delivery date], DAY))

- **Late Delivery Rate:**

Late Delivery % = DIVIDE(CALCULATE(COUNT(Supplychain_DB[Order Id]), Supplychain_DB[Late_delivery_risk]=1), COUNT(Supplychain_DB[Order Id]))

- **Shipping Delay Variance:**

Shipping delay Variance = AVERAGE(Supplychain_DB[Days for shipping (real)]) - AVERAGE(Supplychain_DB[Days for shipment (scheduled)])

- **Delivery Target (Benchmark):**

  Delivery Target = 4

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8. Analysis & Visualizations

This project is divided into three distinct analytical layers, each designed to provide stakeholders with specific levels of detail and actionable intelligence.

<u>Layer 1: Executive Overview (Financial Performance)</u>

Goal: To monitor the organization's overall financial health and provide a high-level summary of revenue and profitability.

- **KPI Cards** (The "Financial Snapshot"):

  Visual Elements: Displays Gross Sales ($36.78M), Total Profit ($3.97M), and Profit Margin (12%).

  Strategic Insight: Provides an immediate health check for the business. While the 12% margin indicates stability, it highlights opportunities for profitability optimization.

- **Sales & Late Delivery Trends (Combo Chart)**:

  Visual Elements: Monthly Revenue (Bars) overlaid with Late Delivery % (Line).

Strategic Insight: Reveals that while revenue remains consistent (averaging $2M–$3M monthly), delivery risk is volatile, specifically peaking during high-volume periods like August.

- **Revenue Distribution (Donut Chart):**

  Visual Elements: Breakdown of Total Sales by Customer Segment (Consumer, Corporate, Home Office).

  Strategic Insight: Identifies the Consumer segment as the primary driver of over 50% of total revenue, marking it as the most critical area for supply chain optimization.

<u>Layer 2: Operational Efficiency (The "Process Investigation ")</u>

Goal: To perform a diagnostic "Deep Dive" into supply chain bottlenecks and identify where operational processes are failing**.**

- **Delivery Speed Gauge (Target vs. Actual):**

  Visual Elements: Average Delivery Days (Actual: 7 Days) vs. Delivery Target (Goal: 4 Days).

  Strategic Insight: Visually quantifies the operational lag. The logistics department is currently underperforming against its benchmark by an average of 3 days.

- **Profit vs. Sales Scatter Chart (Order Leakage):**

  Visual Elements: Total Sales (X-Axis) vs. Benefit per Order (Y-Axis) for every Order ID.

  Strategic Insight: By utilizing a Y-Axis Constant Line at $0, this chart identifies the "Loss Zone." Data points below the line represent "Order Leakage," where shipping costs or excessive discounts have outweighed revenue**.**

- **Shipping Mode Delay Variance (Clustered Bar Chart):**

  Visual Elements: Variance (Actual vs. Scheduled days) across different Shipping Modes.

  Strategic Insight: Isolates carrier unreliability. For example, if "Second Class" shows the highest variance, it signals a need to renegotiate terms with that specific provider.

- **Geographical Risk Map (Bubble Map):**

  Visual Elements: Bubble Size (Total Sales) and Color Gradient (Late Delivery %).

  Strategic Insight: Pinpoints global hotspots. Darker/Red bubbles highlight regions like Western Europe and Central America, where high market success is being undermined by severe delivery delays.

Layer 3: Strategic AI Insights (Root Cause Analysis)

Goal: To move from diagnostic analysis to Prescriptive Analytics by using Machine Learning to identify the hidden drivers of inefficiency.

- **Decomposition Tree (The "Drill-Down" Visual):**

  Function: Interactively breaks down the "Late Delivery %" across multiple dimensions (Region, Category, and Shipping Mode).

  Strategic Insight: Allows to trace delays to their source. Analysis reveals that Europe has the highest delay rate, largely driven by the Technology category.

- **Key Influencers (The "Driver" Visual):**

  Function: Uses AI to determine which variables have the highest statistical impact on late deliveries.

  Strategic Insight: Discovered that selecting 'First Class' shipping increases the likelihood of a late delivery by 1.8x. This identifies the shipping mode itself as a primary bottleneck rather than just the distance.

- **Smart Narrative (The "Automated Storyteller"):**

  Function: Automatically generates natural language summaries of the data findings.

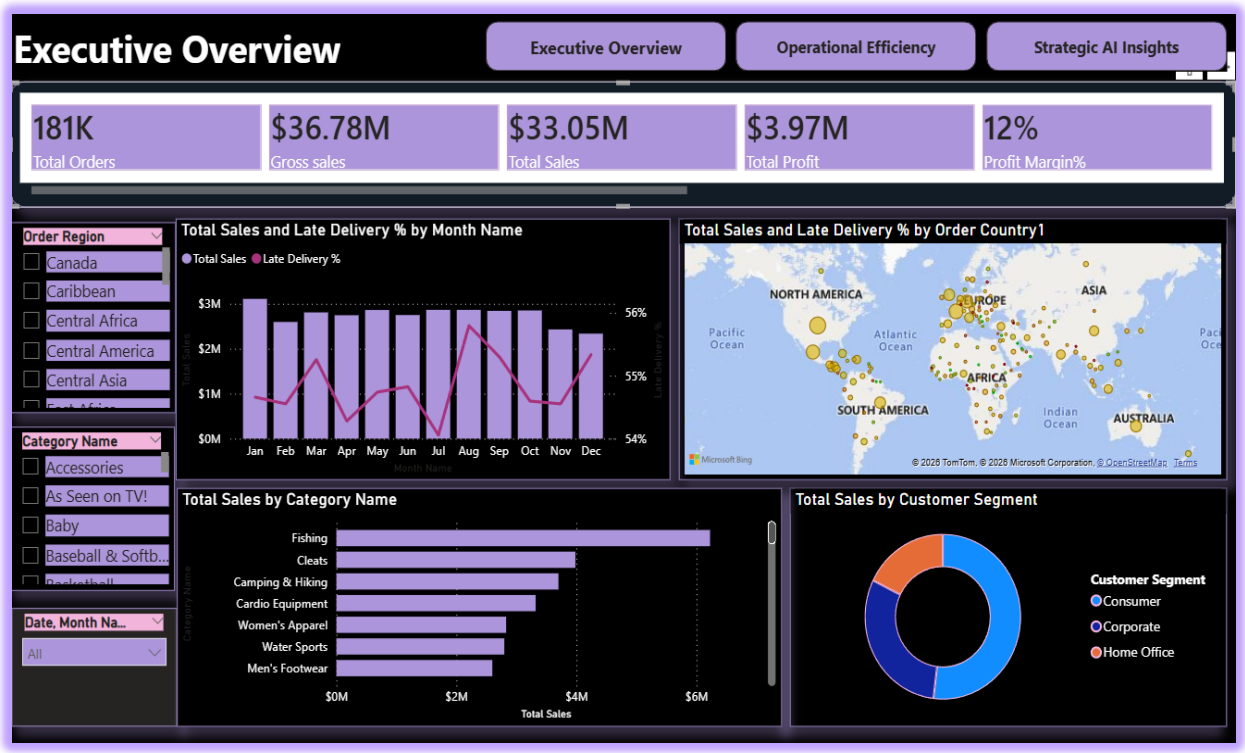
  Strategic Insight: Translates complex AI correlations into plain English, ensuring that non-technical executives can immediately grasp the "bottom-line" findings and take action.

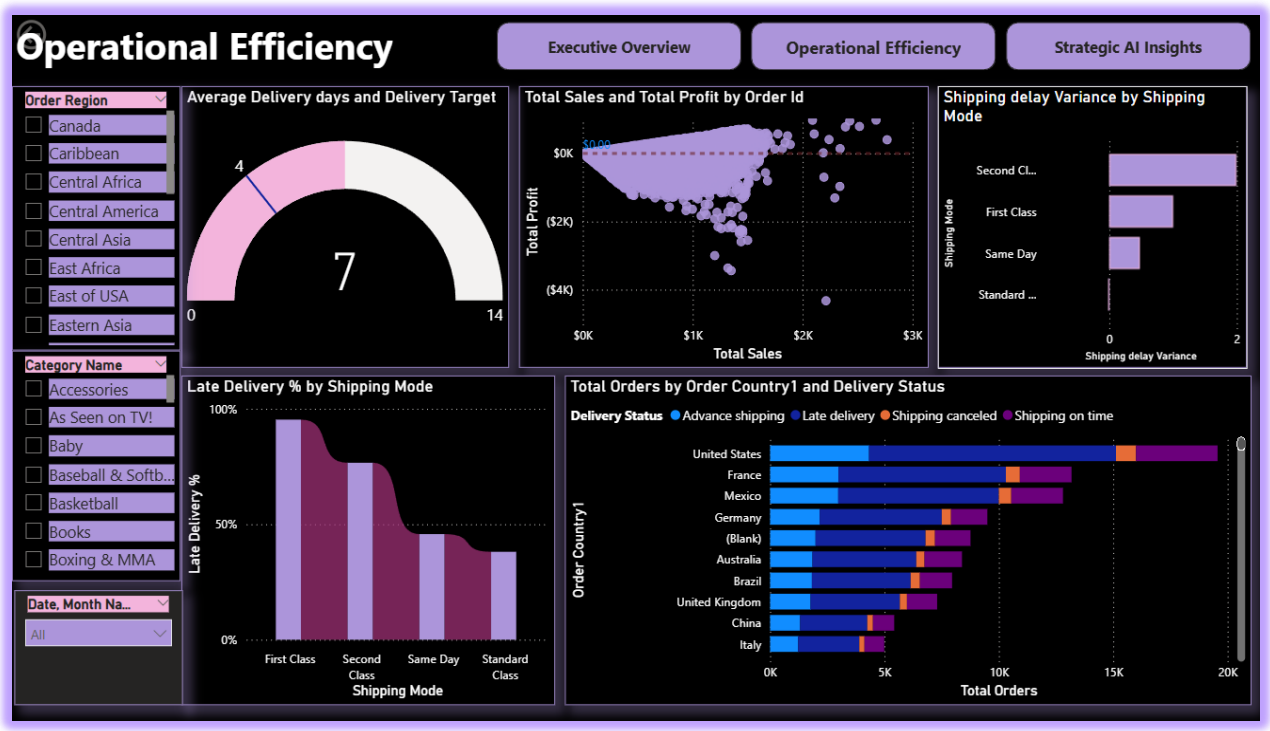
- **Audit Table (The "Granular Detail"):**

  Function: A detailed list of the "Bottom 10" least profitable orders.

  Strategic Insight: Bridges the gap between high-level AI insights and daily operations, allowing the logistics team to audit and resolve specific problematic transactions.

Supply Chain Performance





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9. Insights & Findings

The comprehensive analysis of the DataCo Smart Supply Chain reveals critical operational and financial patterns, categorized by the depth of analytical impact:

9.1. <u>Descriptive & Diagnostic Insights (Operational Performance)</u>

- **Operational Lag:** The supply chain currently operates with a 55% Late Delivery Risk. The actual average delivery time is 7 days, which significantly exceeds the corporate benchmark of 4 days, indicating a 75% performance gap**.**
- **Revenue Concentration:** The Consumer segment is the primary business driver, contributing over 50% of total revenue. This makes the fulfillment efficiency of this segment vital for long-term customer retention.
- **Regional Performance Hotspots:** Western Europe and Central America are identified as "High-Risk" zones. These regions demonstrate high sales volume but simultaneously suffer from the highest frequency of delivery delays.

9.2. AI-Driven & Prescriptive Insights (Root Cause Analysis)

- **Shipping Mode Impact:** AI analysis through Key Influencers reveals that selecting 'First Class' shipping increases the probability of a late delivery by 1.8x. This identifies a major bottleneck in the priority processing phase rather than the transit phase itself.
- **Root Cause Mapping:** The Decomposition Tree traces over 60% of all logistical delays back to the Technology category and Standard/First Class modes within the European and LATAM markets**.**
- **Financial Leakage:** A deep dive into the "Loss Zone" (Bottom 10 Orders) confirms a significant correlation between late deliveries and negative profit margins. This suggests that the cost of expedited shipping and excessive discounting is eroding the bottom line.

9.3.Strategic Recommendations (The Action Plan)

- Based on the data-driven insights above, the following strategic interventions are recommended to optimize the supply chain:
- **Logistics Optimization:** To mitigate high-risk delivery failures, the organization should review carrier contracts and transition high-risk 'First Class' routes in LATAM to more reliable local carriers**.**
- **Operational Audit:** A review of warehouse dispatch priority is required for high-value items in the 'Technology' category to ensure they meet scheduled delivery timelines.
- **Predictive Risk Management:** Implement a Proactive Alert System within the supply chain workflow to flag orders falling into "High-Risk" influencer categories before they exit the warehouse.
- **Profit Margin Protection:** Adjust discount policies for the Consumer segment to ensure that net revenue remains positive even when shipping variances occur.

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10. Conclusion

This project successfully demonstrates the transition from traditional, static reporting to **AI-driven prescriptive analytics** using the DataCo Smart Supply Chain dataset. By integrating comprehensive data cleaning in Excel with advanced relational modeling and DAX computations in Power BI, the following key outcomes were achieved:

- **Visibility Gap Closure:** The development of a professional, interactive 3D dashboard has successfully closed the "Visibility Gap." The organization now has real-time access to logistical bottlenecks, moving away from fragmented, manual reporting and enabling proactive decision-making.

- **Strategic Process Identification:** A critical finding of this study is the identification of the "Shipping Paradox." The analysis revealed that the **"First Class"** shipping mode—intended for speed—is ironically a primary driver of delays (increasing risk by **1.8x**). This insight necessitates a strategic review of carrier contracts and warehouse dispatching protocols.
- **Decision Support & Roadmap:** The final analytical solution provides management with a data-driven roadmap. By identifying the **3-day operational lag** and pinpointing **"Loss Zones"** in the Consumer segment, the dashboard serves as a precise tool for reducing costs and protecting profit margins in high-volume global markets.
- **Transformation of Intelligence:** In conclusion, the application of advanced AI visuals, such as **Key Influencers** and **Decomposition Trees**, has successfully transformed raw transactional data into actionable intelligence. This project proves that data-driven interventions and modern BI tools can significantly optimize global supply chain efficiency and operational reliability.